

PDE

① Classification of PDE $\rightarrow$ General PDE of second order in Two variable $\rightarrow$

$$Rr + Sst + Tt + f(x, y, z, p, q) = 0$$

where R, S, T are coeff. funcn and $r = \frac{\partial^2 u}{\partial x^2}$, $s = \frac{\partial^2 u}{\partial x \partial y}$, $t = \frac{\partial^2 u}{\partial y^2}$ then PDE is said to be

① Hyperbolic at a point (x, y) in D if $S^2 - 4RT > 0$

② Parabolic " " " in D if $S^2 - 4RT = 0$

③ Elliptic " " " if $S^2 - 4RT < 0$

Q $2u_{xx} + 4u_{xy} + 3u_{yy} = 2$
 $R=2, S=4, T=3$

$$S^2 - 4RT \rightarrow 16 - 24 < 0$$

$\therefore$ PDE is elliptic

Q $u_{xx} - (x^2 - y^2)s - u_{xt} + py - q = 2(x^2 - y^2)$
 $R \quad S \quad T$

$$S^2 - 4RT \rightarrow (x^2 - y^2)^2 - 4(xy)(-xy)$$

$$(x^2 - y^2)^2 + 4x^2y^2$$

$$x^4 + y^4 - 2x^2y^2 + 4x^2y^2$$

$$x^4 + y^4 + 2x^2y^2 = (x^2 + y^2)^2 > 0$$

$\therefore$ Hyperbolic

Exmp $\rightarrow$ ① $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$ (1-D Eqn)

$\rightarrow$ wave eqn

$$\left. \begin{array}{l} A=1=R \\ B=0=S \\ C=-1=T \end{array} \right\} \begin{array}{l} 4 > 0 \\ \text{Hyperbolic} \end{array}$$

② $\frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial y}$ (1-D heat Eqn)

Always Parabolic

⊛ Variable Separation →

Q Solve $\frac{\partial u}{\partial z} = 2\frac{\partial u}{\partial x} + u$ by the method of sep of var, where $u(x,0) = 6e^{-3x}$

u depends on x & $T \rightarrow u(x,t) = X(x) \cdot T(t)$

$$\frac{\partial u}{\partial z} = X T' \quad \text{and} \quad \frac{\partial u}{\partial x} = X' T$$

$$\rightarrow X' T = 2 X T' + X T$$

$$2 X T' = T (X' - X)$$

$$\Rightarrow \left(\frac{X' - X}{X} \right) = \frac{2 T'}{T} = k \rightarrow \frac{X' - X}{X} = k \Rightarrow X' - X = kX$$

$$\therefore \boxed{X = e^{(k+1)x}} \leftarrow \boxed{m = k+1}$$

$$\rightarrow \frac{2 T'}{T} = k \rightarrow T' = \frac{k T}{2} \Rightarrow T - \frac{k T}{2} = 0 \rightarrow \boxed{m = \frac{k}{2}}$$

$$\therefore u(x,t) = C_1 e^{(k+1)x} e^{\left(\frac{k}{2}\right)t}$$

$$= C e^{(k+1)x + \frac{k}{2}t}, \quad u(x,0) = 6e^{-3x}$$

$$C e^{(k+1)x} = 6e^{-3x}$$

$$\boxed{C=6} \quad \& \quad k+1=-3 \rightarrow \boxed{k=-4}$$

$$\therefore u(x,t) = 6e^{-2x-2t}$$

Q Solve $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0 \rightarrow u(x,y) = X(x) Y(y)$

$$\rightarrow \frac{\partial u}{\partial x} = X' Y, \quad \frac{\partial u}{\partial y} = X Y' \quad \left\{ \begin{array}{l} X'' Y - 2 X' Y + X Y' = 0 \\ X'' Y - 2 X' Y = -X Y' \\ (X'' - 2 X') Y = -X Y' \end{array} \right.$$

$$\rightarrow \frac{X'' - 2 X'}{X} = \frac{-Y'}{Y} = k$$

$$\rightarrow \frac{X'' - 2 X'}{X} = k \quad \& \quad \frac{-Y'}{Y} = k$$

$$\Rightarrow X'' - 2 X' = X k = 0 \quad \& \quad -Y' = Y k \rightarrow \boxed{Y = C_1 e^{-k y}}$$

$$[m^2 - 2m - k = 0]$$

$$\rightarrow m = 1 \pm \sqrt{1+k} \Rightarrow \boxed{X = C_2 e^{(1+\sqrt{1+k})x} + C_3 e^{(1-\sqrt{1+k})x}}$$

$$u = [C_1 e^{-ky}] [C_2 e^{(\frac{1+\sqrt{1+k^2}}{2})x} + C_3 e^{(\frac{1-\sqrt{1+k^2}}{2})x}]$$

$$Ae^{-ky} = (Ae^{(\frac{1+\sqrt{1+k^2}}{2})x} + Be^{(\frac{1-\sqrt{1+k^2}}{2})x}) e^{-ky}$$

$$\therefore C_1 C_2 = A \quad \& \quad C_1 C_3 = B$$

④ Solving Heat Eqn $\rightarrow$

$$\text{Eqn} \Rightarrow \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\text{Bond Cond} \Rightarrow u(0, t) = 0 = u(l, t)$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$

$$u(0, t) = 0 \Rightarrow u(l, t) = 0$$